

Project Design Phase Solution Architecture

Date	25 July 2026
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Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between enterprise business challenges and technology implementations. For this project, the architecture defines how ServiceNow modules—Service Catalog, Flow Designer, Service Level Management, and CMDB—interact to deliver an automated, error-free network request management solution.

Primary Architectural Goals:

1. **Find the Best Tech Solution:** Replaces disjointed email-based network requests with native ServiceNow SaaS components.
2. **Describe System Structure & Behavior:** Defines end-to-end data progression from portal submission to approval routing and fulfillment execution.
3. **Define Requirements & Specifications:** Enforces strict IPv4 input validation, automated multi-tier approval rules, and automatic catalog task creation.
4. **Deliver Scalable Integration:** Prepares the platform for future zero-touch automated device provisioning via IntegrationHub spokes.

Architectural Components & Specifications

1. Presentation & Data Ingestion Layer (Service Portal)

- **Component:** Service Catalog Item (sc_cat_item).
- **Specification:** Utilizes Catalog UI Policies and Client Scripts for dynamic field visibility (request_type, ip_address_range, access_level) and regex-based IPv4 pattern validation, eliminating invalid inputs at submission.

2. Workflow Orchestration & Business Logic Layer (Flow Designer)

- **Component:** Flow Designer Engine & sysapproval_approver.
- **Specification:** Evaluates request risk upon submission. Standard access routes to the Requester's Line Manager, while administrative access escalates to the Network Security Approvers group. Generates dynamic email notifications with embedded one-click approval buttons.

3. Task Fulfillment & Operations Layer

- **Component:** Catalog Task Engine (sc_task)[cite: 3].
- **Specification:** Automatically creates and assigns fulfillment tasks to the Network Operations Group upon approval[cite: 3]. Mapped form variables display directly in the engineer's task view for immediate context[cite: 3].

4. Asset Integrity & Governance Layer (CMDB & SLAs)

- **Component:** Configuration Management Database (cmdb_ci_netgear) & Service Level Management[cite: 3].
- **Specification:** Automatically updates target device configuration records upon task closure[cite: 3]. Attaches resolution SLAs (8 hours for High/Critical; 24 hours for Standard) with automated breach escalation rules[cite: 3].

ServiceNow End-to-End Solution Architecture

